Haojie Ye

AI Systems and GPU Architecture

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AI performance architect with a Ph.D. in Computer Science and Engineering and a research focus on computer architecture. Connects generative AI workloads to GPU design and product decisions through validated performance models, agent-assisted analysis tools, and investigation of compute, memory, and communication bottlenecks.

Experience

NVIDIA Oct 2024 – Present
Deep Learning Performance Architect Santa Clara, CA

- Led performance modeling studies for LLM inference and video generation, translating model assumptions and execution strategies into GPU bottleneck analysis and product tradeoff recommendations.
- Implemented and validated simulation capabilities for evolving AI workloads, including sparse attention and parallel execution. Checked model shapes, execution mappings, and run assumptions to strengthen performance comparisons.
- Independently developed an agent-assisted performance-attribution component that reconciles execution mappings, explains differences between projections, and surfaces outliers. Integrated it into a broader analysis workflow for repeatable comparison and targeted debugging.
- Evaluated compute, memory, interconnect, and hardware-software tradeoffs for future GPU studies; distinguished workload-specific gains from end-to-end effects to inform engineering and product decisions.
- Presented technical findings across architecture and product teams, developed study materials, and taught colleagues how to run and explain performance modeling workflows.

NVIDIA May 2023 – Aug 2023
Deep Learning Performance Architect Intern Santa Clara, CA

- Researched performance modeling of large language models on future GPU architectures.

Micron Technology May 2022 – Aug 2022
Advanced Hardware Development Intern Allen, TX

- Researched enhanced in-memory computing functions and memory-system design.

Education

University of Michigan, Ann Arbor
Ph.D., Computer Science and Engineering 2021 – 2024
Advisor: Prof. Trevor Mudge. GPA: 4.0/4.0.
M.S.E., Computer Science and Engineering 2019 – 2021
B.S.E., Computer Engineering 2017 – 2019

Shanghai Jiao Tong University
B.S.E., Electrical and Computer Engineering 2015 – 2019

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Selected Research

Hardware-software co-design for recommendation inference, GPU and memory systems, graph processing, and sparse computation. Publications at HPCA, ASPLOS, MICRO, ISCA, VLDB, PACT, and other systems venues. Full publication list on Google Scholar.

Palermo: Improving the Performance of Oblivious Memory using Protocol-Hardware Co-Design
H. Ye, Y. Xia, Y. Chen, K.-Y. Chen, Y. Yuan, S. Deng, B. Kasikci, T. Mudge, N. Talati
HPCA 2025

Protocol and hardware co-design to overlap oblivious-memory operations while preserving security guarantees.

GRACE: A Scalable Graph-Based Approach to Accelerating Recommendation Model Inference
H. Ye, S. Vedula, Y. Chen, Y. Yang, A. Bronstein, R. Dreslinski, T. Mudge, N. Talati
ASPLOS 2023

Graph clustering and embedding partial-sum caching to reduce recommendation inference memory traffic.

Understanding the Performance and Estimating the Cost of LLM Fine-Tuning
Y. Xia, J. Kim, Y. Chen, **H. Ye**, S. Kundu, C. Hao, N. Talati
IISWC 2024

Locality-aware Optimizations for Improving Remote Memory Latency in Multi-GPU Systems
L. Belayneh, **H. Ye**, K.-Y. Chen, D. Blaauw, T. Mudge, R. Dreslinski, N. Talati
PACT 2022

Everest: GPU-Accelerated System For Mining Temporal Motifs
Y. Yuan, **H. Ye**, S. Vedula, W. Kaza, N. Talati
PVLDB 17(2), 2023 / VLDB 2024

NDMiner: Accelerating Graph Pattern Mining Using Near Data Processing
N. Talati, **H. Ye**, Y. Yang, L. Belayneh, K.-Y. Chen, D. Blaauw, T. Mudge, R. Dreslinski
ISCA 2022

Technical Skills

Languages: Python, C/C++, CUDA, Verilog, SystemVerilog.
Performance and architecture: Workload characterization, performance modeling, bottleneck analysis, GPU and memory systems, parallel computing, hardware-software co-design, agent-assisted analysis.
Tools: Git; gem5, gem5-gpu, DRAMSim, Ramulator, CACTI, zsim; Cadence, Design Compiler.

Teaching

University of Michigan — Graduate Student Instructor Sep 2020 – Jan 2021
EECS 598, Applied Parallel Programming with GPUs. Prepared lecture slides, homework, and exams on GPU microarchitecture and CUDA programming with Prof. Reetuparna Das.

Honors

Outstanding Undergraduate Research Award (2019); James B. Angell Scholar (2019).